

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY
Bachelor of Engineering - SEMESTER - II EXAMINATION - SUMMER 2025

Subject Code: BE02000061

Date: 19-06-2025

Subject Name: Industrial Safety and Standards

Time: 10:30 AM TO 01:00 PM

Total Marks: 70

Instructions

- 1. Attempt all questions.**
- 2. Make suitable assumptions wherever necessary.**
- 3. Figures to the right indicate full marks.**

	Marks
Q.1 (a) Define the role of management in safety.	03
(b) Explain fatigue wear.	04
(c) Discuss about the overhauling of electric motor and common troubles involved in it with remedies.	07
Q.2 (a) Explain the role of lubricant.	03
(b) Define Frequency rate, Severity rate, Incidence rate and Activity rate.	04
(c) Describe the structure of hazard identification and risk assessment.	07
OR	
(c) Define fire and explain in details about classification of fire.	07
Q.3 (a) Give a classification of wear.	03
(b) Draw the decision tree for fault tracing in internal combustion engine.	04
(c) List out the types and applications of tools for maintenance.	07
OR	
(a) List the methods for reduction of wear.	03
(b) Draw the decision tree to identify fault in electric motor.	04
(c) Define maintenance and explain different types of maintenance.	07
Q.4 (a) Explain fault tracing.	03
(b) Explain the steps and procedure for periodic & preventive maintenance of pump.	04
(c) Explain in detail about mechanical hazards.	07
OR	
(a) Draw a flow chart of fault-finding sequence.	03
(b) Explain the steps and procedure for periodic & preventive maintenance of air compressor.	04
(c) Explain in details of various causes for Industrial accidents.	07

- Q.5** (a) Enlist the benefits of overhauling. 03
- (b) Explain different types of methods of lubrication. 04
- (c) Explain decision tree and draw the steps involved in it with suitable sketch. 07

OR

- (a) Define periodic maintenance. 03
- (b) Explain in detail about Corrosion and its prevention methods. 04
- (c) Explain in detail about the fault tracing analysis and its importance. 07

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE02000061

Date: 03-01-2026

Subject Name: Industrial Safety and Standards

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Explain in details of various types of Industrial accidents.	03
(b) Explain incidence rate and severity rate for Monitoring Safety Performance.	04
(c) Justify the requirement for Safety policy and safety standards for Industry and organization.	07
Q.2 (a) List the types of hazards.	03
(b) Write a short note on “salient points of factories act 1948 for health and safety”.	04
(c) Explain Fire and explosion hazard rating of process plants.	07
OR	
(c) Explain the structure of hazard identification and risk assessment in detail.	07
Q.3 (a) Write short note on “Service life of equipment”.	03
(b) Explain types and applications of tools used for maintenance.	04
(c) Explain all responsibility of maintenance department.	07
OR	
(a) List all wear reduction methods.	03
(b) Explain type of wear and its cause.	04
(c) List and explain any one corrosion prevention method.	07
Q.4 (a) Classify lubricants in details.	03
(b) Explain all factors affecting the corrosion.	04
(c) Explain decision tree concept and its applications.	07
OR	
(a) Draw a flow chart of fault-finding sequence.	03
(b) Write short note on” Fault tracing and its importance”.	04
(c) Explain different types of faults in machine tools and its causes.	07

- Q.5 (a)** Differentiate between Periodic and preventive maintenance. **03**
- (b)** Explain Steps for periodic and preventive maintenance of electrical motor. **04**
- (c)** Explain Repair cycle concept and importance. **07**

OR

- (a)** Identify different advantages of preventive maintenance. **03**
- (b)** Explain Steps for periodic and preventive maintenance of Diesel generating (DG) sets. **04**
- (c)** Explain overhauling of mechanical components in details. **07**
